

03-advanced-troubleshooting

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Advanced Troubleshooting

Audience: Admin operators diagnosing problems that require Apps Script editor access, Script Properties access, or log inspection. If the issue is something a manager can resolve from the spreadsheet UI, it belongs in </docs/waratah/for-managers/05-troubleshooting.md>, not here.

This file holds the diagnostic procedures and recovery paths that go deeper than the manager troubleshooting guide.

1. Where to Start

When a problem reaches you, work through this decision tree before diving into specific sections:

1. **What is the user-visible symptom?** (Slack message missing, email missing, rollover didn't run, task DM not received, dashboard showing stale numbers.)
2. **When did it last work?** (Yesterday, last week, this morning?)
3. **Has anything changed recently?** (A clasp push, a Script Properties edit, a new staff member added, a Slack workspace change?)
4. **Is the issue affecting one user, one feature, or the whole system?**

The combination usually points at the right section below. If not, start with the logs (Section 2).

2. How to View Logs

Apps Script has two logging surfaces. Both are essential.

Executions panel

Open the relevant Apps Script project (Shift Report or Task Management) > left sidebar >

Executions. This shows every script execution in the last 30 days with: function name, trigger type (manual, time-based, on edit), start time, duration, status (succeeded, failed, timed out).

Filter by status = Failed to see only failures. Click any execution to see its full log output.

Logger output (within an execution)

Inside an execution's detail view, the **Logs** tab shows everything the script wrote via `Logger.log(...)`. This is the primary debugging output.

What to look for

- **Cannot call SpreadsheetApp.getUi() from this context** : a UI call ran in a trigger context (where there is no UI). Code bug; report to developer (Phase 4).
 - **Exception: You do not have permission to call ...** : the script lost permission to a resource (Drive folder, sheet). Re-authorise the script (Run any function from the Apps Script editor; you will be prompted to re-grant permissions).
 - **Service Spreadsheets failed** : a transient Google Sheets API error. Usually a one-off. Re-run the function.
 - **Quota exceeded** : the script hit a Google quota (daily email send, daily UrlFetch, etc.). Wait until midnight Pacific time and retry, or contact Google Workspace admin.
 - **Silent failures with no log**: the function did not run at all. Check the Triggers panel (Section 8).
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3. Slack Webhook Diagnostics

Symptom: Slack message did not post

Check in this order:

1. **Did the rest of the send succeed?** Email and warehouse should still have worked. If yes, the failure is isolated to Slack.
2. **Is Slack itself working?** Check the Slack workspace status; check whether other messages are appearing in the manager channels.
3. **Is the webhook URL valid?** Test it with a curl command from the terminal:

```
""bash curl -X POST -H 'Content-type: application/json' \ --data '{"text":"Webhook test"}' \
'https://hooks.slack.com/services/...' "" - HTTP 200 with body ok : the webhook works. The
problem is in the script. - HTTP 404 or body no_service / invalid_token : the webhook is expired
or revoked. Regenerate in Slack. - HTTP 410 channel_not_found : the target channel was renamed
or archived. Regenerate.
```

Symptom: One person's DM not arriving

The personal webhooks live inside `SLACK_DM_WEBHOOKS` (JSON object). Check:

1. Is their name spelled exactly as in `STAFF_LIST` (case-sensitive)? "Joffy" not "joffy".
 2. Is the JSON syntactically valid? A missing comma breaks the whole object. Paste it into a JSON validator to confirm.
 3. Is the webhook URL still valid? Test with curl as above.
 4. Is the property updated in BOTH Apps Script projects? See [02-staff-and-access-management.md](#) Section 8 for the dual-update rule.
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4. Email Recipient Diagnostics

Symptom: One recipient is not receiving emails

In order:

1. **Check their spam folder.** Search "Waratah shift report".
2. **Send a TEST email manually.** Run `verifyScriptProperties()` to confirm the recipient is in the list. Then send a TEST shift report and check the Executions panel for `MailApp.sendEmail` calls and their addressee.
3. **Check the JSON object validity.** `WARATAH_EMAIL_RECIPIENTS` must be a valid JSON object mapping email to display name. Stray commas, mismatched quotes, or duplicate keys break the parse.
4. **Check Gmail daily send quota.** If the script has sent more than 100 emails in 24 hours (Workspace) or 20 emails (free Gmail), MailApp will fail. Wait 24 hours or upgrade the account.
5. **Check whether their mailbox is full.** Ask the recipient.

Symptom: All recipients are missing emails

If no recipient received the email, the failure is at the script level, not per-recipient:

1. Check Executions panel for the relevant nightly send. Look at the log output for `MailApp.sendEmail` calls.
2. If `MailApp.sendEmail` did not appear in logs, the email step did not run. The send pipeline failed before reaching email; check the warehouse and Slack steps.
3. If `MailApp.sendEmail` appeared but logged an error: read the error. Common: quota exceeded, sender address not authorised.

5. Rollover Recovery

The weekly rollover (Monday 9pm) has the most moving parts. When it fails partway, the spreadsheet can be in an inconsistent state: tabs renamed but cells not cleared, or PDF saved but data not cleared, and so on.

Symptom: Tuesday morning, the tabs still show last week's dates

The rollover did not run, or ran and failed before the rename step.

Recovery:

1. Check Executions panel of the Shift Report project for Monday around 9pm. Look for `runWaratahWeeklyRollover` (current) or `performWeeklyRollover` (legacy name).
2. If no execution exists, the trigger did not fire. Check the Triggers panel: is the trigger installed?
3. If the execution failed, read the log to see where it failed.
4. Recovery: from the spreadsheet menu, run **Waratah Tools > Admin Tools > Weekly Reports > Weekly Rollover (In-Place) > Run Rollover Now** (password gated). This re-runs the rollover

idempotently. If the previous run had partial success, this one should complete the unfinished steps.

Symptom: Tabs are renamed for new week, but old data is still in cells

The rollover archived and renamed but the clear step did not complete.

Recovery:

1. Do not let anyone type fresh data into the cells.
2. Confirm with the Executions panel: did `runWaratahWeeklyRollover` complete fully or fail at the clear step?
3. Re-run the rollover idempotently from **Waratah Tools > Admin Tools > Weekly Reports > Weekly Rollover (In-Place) > Run Rollover Now**. The rollover is designed to complete unfinished steps when re-run, so this will normally clear the lingering data.
4. If a re-run does not clear the cells, contact the developer for an ad-hoc cleanup; there is no dedicated "Clear Manager Inputs" menu item.
5. Verify the cells are now empty.

Symptom: PDF or spreadsheet copy missing from Drive archive

Drive permission failure, or the `ARCHIVE_ROOT_FOLDER_ID` is pointing at the wrong folder.

Recovery:

1. Check the Drive folder ID in `ARCHIVE_ROOT_FOLDER_ID` Script Property. Open that ID in Drive and confirm it is the expected archive folder.
2. Confirm the script owner has Edit access to the folder.
3. Re-run **Waratah Tools > Admin Tools > Weekly Reports > Weekly Rollover (In-Place) > Run Rollover Now** which re-archives the week.

Symptom: Rollover ran but Slack confirmation did not post

The rollover succeeded; the confirmation post failed. This is a Slack webhook issue. See Section 3.

6. Trigger Destruction After Deployment

When you push code via `clasp push`, **Google Apps Script may destroy all installed time-based triggers** for the project. This is by design (the new code may have renamed or removed the trigger handler functions, so old triggers would orphan).

After every clasp push, **assume triggers need re-installation**. See [04-deployment-and-clasp.md](#) for the full post-deploy checklist.

How to detect trigger destruction

1. Open Apps Script project > left sidebar > **Triggers**.
2. Count the triggers. The expected count:

- Shift Report project: 3 time-based triggers (rollover Mon 9pm, backfill Mon 8am, digest Mon 4pm). - Task Management project: up to 6 installable triggers (bi-hourly cleanup, daily 6am workload, daily 6am task maintenance, Mon 6am archive, Mon 10am summary, on-edit handler).

1. If a count is short, re-install the missing triggers from the relevant menu items.

Shift Report project triggers

Handler	Schedule	Menu path to install
runWaratahWeeklyRollover	Mon 9pm	Waratah Tools > Admin Tools > Setup & Utilities > Setup All SR Triggers OR Waratah Tools > Admin Tools > Weekly Reports > Weekly Rollover (In-Place) > Create Rollover Trigger (Mon 9pm)
runWeeklyBackfill_	Mon 8am	Setup All SR Triggers OR Waratah Tools > Admin Tools > Data Warehouse > Setup Weekly Backfill Trigger
sendWeeklyRevenueDigest_Waratah	Mon 4pm	Setup All SR Triggers OR Waratah Tools > Admin Tools > Weekly Digest > Setup Monday Digest Trigger

Task Management project triggers

Handler	Schedule	Menu path to install
cleanupAndSortMasterActionables	every 2 hours	Task Management >  Admin Tools >  Setup Triggers > Create Bi-Hourly Cleanup Trigger (Every 2hrs)
runScheduledStaffWorkload	Daily 6am	Task Management >  Admin Tools >  Setup Triggers > Create Daily Staff Workload Trigger (6am)
runDailyTaskMaintenance	Daily 6am (fires within the 6 to 7am window)	Not exposed in the current menu; installable from the Apps Script editor by running <code>createDailyMaintenanceTrigger()</code>
runScheduledArchive	Mon 6am	Task Management >  Admin Tools >  Setup Triggers > Create Weekly Archive Trigger (Mon 6am)
sendWeeklyActiveTasksSummary	Mon 10am	Task Management >  Admin Tools >  Setup Triggers > Create Weekly Summary Trigger (Mon 10am)
onTaskSheetEditWithAutoSort	onEdit (installable, not simple)	Task Management >  Admin Tools >  Setup Triggers > Create Edit Trigger (Auto-sort)

After installing, verify each trigger appears in the Triggers panel with the correct function name and schedule.

7. Revenue Digest Diagnostics

Symptom: No digest posted Monday at 4pm

1. Executions panel: did `sendWeeklyRevenueDigest_Waratah` run Monday at 4pm? If not, trigger not installed; see Section 6.
2. If it ran and failed: read the log. Common: warehouse spreadsheet not reachable (`WARATAH_DATA_WAREHOUSE_ID` Script Property is wrong or revoked), or no data rows found for the week (warehouse never received the week's nightly data; check the backfill).
3. Recovery: run **Waratah Tools > Admin Tools > Weekly Digest > Send Revenue Digest (LIVE)** to manually post the digest.

Symptom: Digest posted but shows zero or negative revenue

The data is in the warehouse but malformed:

1. Open the warehouse spreadsheet (`WARATAH_DATA_WAREHOUSE_ID`).
 2. Filter `NIGHTLY_FINANCIAL` by date range = last week.
 3. Inspect each row's Net Revenue column. Look for negatives, blanks, or extreme outliers.
 4. If a row has a typo (e.g. cash variance accidentally typed as negative net revenue), correct it manually in the warehouse.
 5. Re-run the digest with **Waratah Tools > Admin Tools > Weekly Digest > Send Revenue Digest (LIVE)** to repost with the corrected number.
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8. Dashboard Rebuild

The Analytics and Executive Dashboard tabs in the shift report spreadsheet rebuild automatically on a schedule, but you can force a rebuild.

When to force rebuild

- After a manual warehouse correction (Section 7 above) to refresh the dashboards.
- After deploying new code that changed the dashboard formulas.
- When you spot stale or wrong numbers and want to ensure the dashboard is computing fresh.

Procedure

1. Open the shift report spreadsheet.
2. **Waratah Tools > Admin Tools > Analytics > Build Financial Dashboard** (password gated).
3. Then **Waratah Tools > Admin Tools > Analytics > Build Executive Dashboard** (password gated).

4. Wait for the success messages; each can take 30 to 60 seconds.
5. Verify the dashboards show the expected current values.

If rebuild fails, check the Executions panel for the function `pw_buildFinancialDashboard` or `pw_buildExecutiveDashboard` and read the error.

9. Task Management Diagnostics

Symptom: Tasks from last night's shift report did not appear in Task Management

1. Check Executions panel of the Shift Report project for the relevant nightly send. Look for the task-sync step (`pushTodosToMasterActionables` runs as part of the export pipeline). Did it run?
2. If it ran and failed: read the error. Common: `TASK_MANAGEMENT_SPREADSHEET_ID` Script Property is wrong, or the Task Management spreadsheet is unreachable.
3. Recovery: re-run the export for that day. `pushTodosToMasterActionables` runs inside `exportAndEmailPDF` , so re-running the export will re-sync the tasks. There is no standalone "Re-sync Tasks" menu item.

Symptom: Task DMs not arriving for assignments

See Section 3 above (Slack webhook diagnostics). Apply the dual-update rule: confirm `SLACK_DM_WEBHOOKS` is set in BOTH projects.

Symptom: BLOCKED task did not escalate after 14 days

The escalation runs as part of the daily 6am task maintenance trigger (fires within the 6 to 7am window).

1. Check Executions panel of the Task Management project for `runDailyTaskMaintenance` on a recent morning. Did it run?
 2. If yes: read its log for the escalation step. The log should list any tasks that triggered escalation.
 3. Confirm `ESCALATION_EMAIL` and `ESCALATION_SLACK_WEBHOOK` Script Properties point at correct targets.
 4. Recovery: from the Task Management spreadsheet menu, **Task Management >  Admin Tools > Manual Actions > Check Blocked Escalations Now.**
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10. Quick-Reference Table

Symptom	First check	Section
Slack message missing	Slack workspace status, then webhook curl test	3
Email missing for one person	Spam folder, then JSON validity	4
Email missing for all	Executions panel, MailApp.sendEmail log	4
Rollover didn't run	Executions panel for <code>runWaratahWeeklyRollover</code>	5
Tabs renamed but data not cleared	Confirm rollover partial-fail state, then clear menu item	5
Trigger missing after deploy	Triggers panel count vs expected (3 SR + 6 TM)	6
No digest Monday afternoon	Trigger installed? Then warehouse reachable?	7
Dashboard showing wrong numbers	Force rebuild	8
Tasks from last night missing in Task Management	Task-sync step in execution log	9
Task DM not arriving	SLACK_DM_WEBHOOKS in both projects	9
BLOCKED task not escalating	Daily task maintenance execution log	9

11. Escalating Beyond Yourself

If you have worked through this guide and the issue is unresolved, you need developer help. Compose a report including:

1. **Symptom in one sentence.** "Friday night's Slack message did not post."
2. **The execution log entry** (copy-paste the failing Executions panel entry, redacting any webhook URLs).
3. **What you tried.** "Tested webhook with curl, returned ok. Resent the report manually, also failed."
4. **What you suspect.** "Possibly a script-side issue, not webhook."

Send to Evan, or if Evan is offline, send to the integration alert email (`INTEGRATION_ALERT_EMAIL_PRIMARY` in Script Properties).

For after-hours weekend service-night failures, call rather than email. The on-call response window is tight.